

University of Windsor
Department of Civil and Environmental Engineering
CIVL 8360 – Earthquake Engineering

Term Project
Posted: August 26, 2025
Dr. N. Van Engelen (niel.vanengelen@uwindsor.ca)

Overview

This project is to examine a selected technical problem or topic within earthquake engineering. The report **must** include a brief literature review **and** an analytical component (e.g., parametric study, time history analysis, model, design). The analytical component does not need to be novel but should demonstrate or reinforce a concept related to the topic.

The material should be presented as explaining and demonstrating the topic to your peers. Your submissions may be shared with your peers.

Group

The number of members in each group, if applicable, will be announced on Brightspace and in class.

The same grade will be awarded to all members in the group. If members do not contribute equally, you may confidentially request that a peer evaluation be completed. All members will be required to submit a confidential peer evaluation. Grades will be adjusted for members who contribute significantly less or more based on the evaluation results at the instructor's discretion. In some cases, the group members will be required to meet with Dr. Van Engelen to discuss individual contributions. In all cases, concerns over group member participation should be brought to Dr. Van Engelen's attention **before** the deadline.

Potential Topics

Your selected topic cannot be an area that you or your partners are currently or have previously conducted research in or submitted a course project on. Suggested topics include (as they relate to earthquake engineering):

- Base isolation (focus on different types of devices)
- Base isolation (multi-level isolation systems)
- Distributed damping systems
- Rocking structures (uncontrolled)
- Controlled rocking systems
- Ground motion prediction equations
- Lateral force resisting systems (select any system)
- Equal displacement assumption
- Backbone curves and push over tests
- Viscous dampers
- MR dampers
- Modelling damping in structures
- Ground motion scaling and selection
- Modelling force-displacement hysteresis loops
- Negative stiffness devices
- Modal combination rules
- Shape memory alloys

You may propose a topic not listed above for review and approval by Dr. Van Engelen. The topics will be awarded on a first-come first-serve basis. No two groups/individuals are permitted to do the same topic (although different aspects of the same topic are permitted).

Requirements

Task 1: Select a Group (0%)

Submit a list of your group members to Dr. Van Engelen.

Submission instructions will be posted on Brightspace.

Task 2: Select a Topic (1%)

Submit a list of topics in order of preference (minimum 3) on Brightspace. If the topic is not on the list above, please provide a short statement to describe the area. Once the topic has been assigned you may begin developing the scope. A change in topic after approval has been received is permitted but must be discussed with and approved by Dr. Van Engelen.

Due by end of day (11:59 pm) on Tuesday, September 23, 2025.

Task 3: Select a Scope (4%)

Develop a scope for your project that clearly identifies and outlines the focus of the project and the analytical component in the form of a short (~1/2 page) written submission. Also include a tentative outline of your report identifying major sections and subsections. Submit this task as a single PDF file on Brightspace.

Due by end of day (11:59 pm) on Tuesday, September 30, 2025.

Task 4: Submit a Progress Update (20%)

Submit a progress update on Brightspace containing these sections:

- a short bullet point list of activities to date;
- a plan outlining how you intend to complete the project;
- any challenges you've encountered; and,
- questions requiring discussion.

You are required to submit a draft of your literature review at this time. All groups will be required to meet with Dr. Van Engelen to discuss their progress.

Due by end of day (11:59 pm) on Tuesday, October 28, 2025.

Task 5: Written Technical Report (45%)

Present your findings as a professional engineering technical report. The maximum length of the report is 18 pages excluding the title page, table of contents, executive summary, appendices and references. Use 11 pt. Times New Roman font with 2.5 cm margins all around, fully justification and 1.5 line spacing. There is no limit on appendices. Supplemental material may be requested by Dr. Van Engelen during marking of the project.

Due by end of day (11:59 pm) on Wednesday, November 26, 2025.

Task 6: Oral Presentation (25%)

Conduct a maximum 10-14-minute oral presentation followed by questions. The presentation should be professional and appropriate for a technical audience. Use of PowerPoint or similar software is required. All group members must participate in the presentation approximately equally. An individual grade of 0 will be awarded if a group member does not participate.

Submit a copy of your slides on Brightspace by 11:59 pm on Wednesday, November 26, 2025.

A presentation schedule will be made available on Brightspace. Presentations during the week of December 1-5, 2025. A special session may be arranged.

Task 7: Oral Presentation Peer Evaluation (5%)

For each presentation, students will be assigned to evaluate the oral presentation. Peer evaluations require justification for the grade in order to be considered (i.e., unjustified high or low grades will be discarded). Students will be evaluated for the quality of their feedback and comments on the presentation. Each peer reviewer is required to ask at least one question during the presentation. The quality of the question will be assessed.

A maximum 1/2 page (11 pt. Times New Roman, 1.5 line spacing) justification must be included with the grade sheet. The evaluation schedule will be posted on Brightspace.

Due end of day (11:59 pm) on the day the presentation is given.

All dates are tentative. Any changes or clarifications will be announced on Brightspace. It is your responsibility to stay up-to-date with the project requirements.

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Term Project Progress Update Rubric

Date: _____

Group: _____

Topic: _____

Evaluator: _____

OVERALL: ____ /20

Progress Update: ____ / 5

Contains all four required sections.

Literature Review Draft: ____ / 15

A complete literature review is provided, clear, and substantially completed.

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Term Project Presentation Rubric

Date: _____ Group: _____

Topic: _____ Evaluator: _____

OVERALL: _____ /25 · **Time Multiplier** _____ = _____ /25

Time Limit:

The presentation was completed within the 10-14-minute duration

Multiplier:

- 0.95 for up to 1 minute outside target duration
- 0.75 for up to 2 minutes outside target duration
- 0.50 for up to 3 minutes outside target duration

Organization: _____ /5

Clear and logical organization and presentation flow; well defined introduction and summary/conclusion; substantial details in body of presentation

Content: _____ / 12

Presentation is useful and appropriate for the audience to learn about the topic; evidence of significant research/depth; level of material is appropriate for a graduate course in civil engineering

Response to Questions: _____ / 3

Questions are well considered and demonstrate a clear understanding of the topic

Presentation Delivery: _____ / 5

Professional behaviour of the presenters; clear voice and enunciation; evidence of rehearsal; overall quality of visual material; suitable use of colour and contrast; readability; appropriate number of figures and visual aids

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Term Project Report Rubric

Group: _____

Topic: _____

OVERALL: ____ /45

Presentation and formatting: ____ / 8

Properly formatted as a professional engineering report; followed formatting requirements; general appearance is professional; figures are high quality; correct captions and in-text referencing; appropriate amount and use of figures and tables

Organization and flow: ____ / 5

Clear and logical flow; use of sections and subsections is appropriate

Content: ____ / 15

Clear introduction and summary/conclusion; sufficient details and scope; demonstrates a clear understanding of the topic; depth is appropriate for a graduate level civil engineering course

Analytical component: ____/10

Contains an analytical component/investigates the topic beyond a literature review/ technical depth is appropriate for a graduate level civil engineering course

Grammar and writing: ____ / 5

Minimal spelling and grammatical errors; uniform language style throughout the report; writing is appropriate for a professional engineering report

Referencing: ____ / 2

All sources are properly cited; references are provided and formatted correctly

Note: Weighting may be adjusted depending on the infraction.